Supplementary Material: Stable Explainability-Guided Sparse Sensor Selection for Subject-Independent Surface Electromyography Hand-Gesture Recognition

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1 Detailed Hyperparameter Search Spaces

Table S1 documents the bounded hyperparameter search ranges utilized for all classifiers.

Table 1: Model Hyperparameter Search Spaces

Model	Hyperparameter	Search Range / Value
XGBoost	n_estimators	250
	max_depth	6
	learning_rate	0.05
	subsample	0.8
	colsample_bytree	0.8
SVM	Kernel	RBF
	C	1.0
	gamma	scale
Random Forest	n_estimators	200
	max_depth	15

2 Sensitivity Analysis: Non-Inferiority Margins

We performed sensitivity analyses at non-inferiority margins $\delta = 0.01$, $\delta = 0.02$ (primary), and $\delta = 0.03$. The minimum non-inferior channel count k^* remained consistent across margins ($k^* = 6$ at $\delta = 0.02$).

3 Quality Control and Dataset Integrity

All 43 GRABMyo participant recordings and 10 NinaPro DB5 subject files passed zero-NaN and non-flat signal checks. Cryptographic SHA-256 manifests are preserved under `data/manifests/`.